



BUTACU IONEL-CATALIN

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SUMMARY

Machine Learning Engineer with a background in software development, bringing a strong academic foundation in deep learning, NLP, and computer vision. Currently pursuing a Master's in Artificial Intelligence, with hands-on experience in model training, feature engineering, and AI application development.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Software Engineer

Iași, RO

June 2023 – Present

- Built features spanning Qt's UI, networking, model/view, and event-handling layers for real time applications.
- Designed scalable software components by applying core OOP principles and industry-standard design patterns.
- Utilized C++ STL containers and algorithms to implement efficient business logic and data processing.
- Decomposed complex requirements into scoped deliverables, consistently meeting sprint estimations.
- Reduced technical debt through targeted refactoring and clean code, improving codebase resilience.
- Diagnosed and resolved production crashes and memory leaks using debugging tools and root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Built UI components from design specifications, integrating them into existing production views.
- Delivered pixel-accurate UI and extended feature parameters through import/export integrations.
- Practised all Git workflow stages: branching, commits, merge requests, code reviews, and CI/CD.

Special Telecommunications Service (STS)

IT Technician – Electoral Bureau

Botoșani, RO

August 2020 – September 2020

- Supported the electoral process by ensuring IT infrastructure was operational at electoral bureaus.
- Assisted in verifying and transmitting electoral data through IT systems (SIMPV/SICPV).
- Offered technical assistance and guidance to electoral bureau staff.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25) [🔗](#)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25) [🔗](#)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25) [🔗](#)

SKILLS & COMPETENCIES

- | | |
|--|--|
| ◦ Feature Engineering & Data Preprocessing | ◦ Model Training & Hyperparameter Tuning |
| ◦ ML Experiment Design & Evaluation | ◦ Object Detection & Image Processing |
| ◦ Cloud Infrastructure & API Development | ◦ LLM Integration & RAG Systems |
| ◦ Vector Databases & Semantic Search | ◦ Model Quantization & Edge Deployment |
| ◦ Software Architecture & Clean Code | ◦ Version Control & CI/CD Pipelines |

EDUCATION

Technical University of Iași

Master of Science in Artificial Intelligence

Iași, RO

October 2024 - June 2026

Advanced Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Systems Engineering

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Data Processing, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Numerical Methods, Neural Networks, Optimization Techniques, Fuzzy Logic, Knowledge-Based Systems

PROJECTS

- GenAI with Physics Constraints (in progress)**  **2026**
Text-to-video pipeline that generates physics-realistic 3D animations from natural language prompts. Integrates Shap-E for 3D asset generation, motion synthesis models, and PyBullet for rigid-body physics simulation, enhanced by ControlNet.
- IDS/IPS Smart Firewall**  **2026**
Collaborative intelligent firewall system featuring a CNN threat classifier and an RL adversarial agent. Personal contribution focused on system architecture, Snort rule configuration, and developing a DBSCAN clustering module that analyzes runtime payloads to auto-generate rules, reducing ML inference load.
- Hand Gesture Recognition**  **2025**
YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). Optimised for NPU/CPU deployment via TFLite INT8 quantisation, CLAHE contrast preprocessing, and temporal smoothing to reduce false positives.
- AI Medication Assistant**  **2025**
Full-stack RAG application: React/TypeScript frontend (GitHub Pages) backed by a Gradio API (Hugging Face Spaces). FAISS-indexed knowledge base of 1200+ medications across 20 medical categories; answers symptom queries and surfaces direct pharmacy links.
- Metaheuristic Framework**  **2025**
Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.
- fMRI Brain-Age Prediction**  **2025**
Competition ML pipeline predicting biological brain age from ~20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.

VOLUNTEER EXPERIENCE

- LOCAL BEST Group Iași** **Iași, RO**
IT Department Coordinator **Sep 2023 – Aug 2024**
- Directed IT operations and maintained technical infrastructure across organisational events.
 - Mentored volunteers and oversaw delivery of technical projects.
 - Administered hardware, software licensing, and network resources.
- Event Coordinator | JobShop Career Event** **Nov 2022 – Apr 2023**
- Coordinated logistics and operations for a large student–industry career event.
 - Managed scheduling, partner engagement, and on-site execution.
 - Improved attendee experience through strategic planning and workflow optimisation.
- TOPICS Responsible | Prove It 24H Techathon** **Dec 2021 – Mar 2022**
- Defined technical requirements and coordinated hackathon logistics.
 - Supported participants with rules, resources, and real-time assistance.
 - Facilitated communication between organisers, industry mentors, and teams.
- IT Responsible | BEST Training Weekend** **Jan 2021 – May 2022**
- Developed and maintained the event website using HTML, CSS, and JavaScript.
 - Collaborated with design and marketing responsible members on content, branding, and technical setup.

ADDITIONAL SKILLS

- Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain&LangGraph, OpenCV
- MLOps & Cloud:** Docker, MLflow, W&B, ONNX, Oracle OCI/OAC, Azure ML
- Data & Dev:** Python, C++, SQL, Spark, Flask, FastAPI, React, NodeJS
- Soft Skills:** Problem Decomposition, Clear Communication, Knowledge Sharing, Self-directed Learning